



PRODUCT DATA SHEET

NEAT CUTTING OIL - TECHNOFLUID NEAT CUTTING OIL 55

Product Description

- • TECHNOFLUID Neat Cutting Oil 55 is a high-performance straight (neat) cutting oil formulated for heavy-duty metal machining operations.
- • Blended from high-quality base oils and fortified with extreme pressure (EP) additives containing active sulphur and chlorine to withstand severe machining conditions.
- • Provides excellent lubricity and load-carrying capacity, ensuring smooth cutting action and superior surface finish.
- • Designed for operations requiring high tool protection, reduced friction, and efficient heat dissipation.
- • Suitable for use where water-based cutting fluids are not recommended.

Key Performance Benefits

- • Excellent extreme pressure properties protect tools during heavy-duty machining.
- • Reduces tool wear, scuffing, and built-up edge formation.
- • Improves surface finish and dimensional accuracy of machined components.
- • Effective heat removal from the cutting zone.
- • Superior wettability ensures deep penetration into the cutting interface.
- • Minimizes tool breakage and extends tool life.
- • Ensures stable performance over extended operating periods.

Applications Recommended for heavy-duty machining operations such as:

- Gear hobbing
- Thread cutting
- Broaching
- Tapping and deep drilling

Suitable for machining of:

- High-tensile steels
- Alloy steels
- Stainless steels
- Shear-resistant and difficult-to-machine materials

Widely used in automotive, engineering, and precision machining industries.

Typical Uses

- Operations demanding high EP performance and superior lubrication.
- Machining applications involving high cutting forces and temperatures.
- Precision components requiring excellent surface finish.

Operating Characteristics

- Medium viscosity ensures effective lubrication and chip removal.
- Stable formulation minimizes oil degradation during continuous use.
- Compatible with common machine tool materials and sealing systems.

Typical Characteristics

Characteristics	Test Method	Typical Figures
Appearance	Visual	Clear & Bright
Colour	ASTM D1500	≤ 4.0
Base Oil Type	—	Mineral
Additive Type	—	EP additives (Sulphur / Chlorine based)
Density @ 15°C, g/ml	ASTM D4052	0.885
Kinematic Viscosity @ 40°C, cSt	ASTM D445	50 – 60
Kinematic Viscosity @ 100°C, cSt	ASTM D445	7.5 – 9.0
Viscosity Index	ASTM D2270	≥ 95
Flash Point (COC), °C	ASTM D92	≥ 200
Pour Point, °C	ASTM D97	-6
Copper Corrosion (3 hrs @ 100°C)	ASTM D130	1b
Four Ball Weld Load, kg	ASTM D2596	≥ 250
Wear Scar Dia, mm	ASTM D2266	≤ 0.50
Oxidation Stability	ASTM D943	Excellent
Rust Protection	IP / In-house	Pass
Active Sulphur	—	Mildly Active
Chlorine Content	—	Present

